

Introduction to Robotics

M.. RAMDANI

University Badji Mokhtar - Annaba
Faculty of Technology
Department of Electronics
PO. Box. 12, Annaba, 23000, Algeria

Cours L3 Automatique 2026/2027



Chapter 1:	Introduction
Chapter 2:	Actuators and Drive Systems
Chapter 3:	Robot Mechanisms
Chapter 4:	Kinematics
Chapter 5:	Dynamics
Chapter 6:	Compliant force control

Advanced Topics

Chapter 7:	Path Planning
Chapter 8:	Probabilities
Chapter 9:	Mobile Robotics
Chapter 10:	Simultaneous Localization and Mapping (SLAM)

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 - Dynamics of single-Axis Drive Systems
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 - Basic 3D geometry
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Chapter 6: Compliant force control: Position and force control and compliance control.

Advanced topics

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Chapter 10: Simultaneous Localization and Mapping (SLAM)

Localization and Mapping: Types of maps: Grid map, Laser scan map, Victoria Park data set

Simultaneous Localization and Mapping Problem

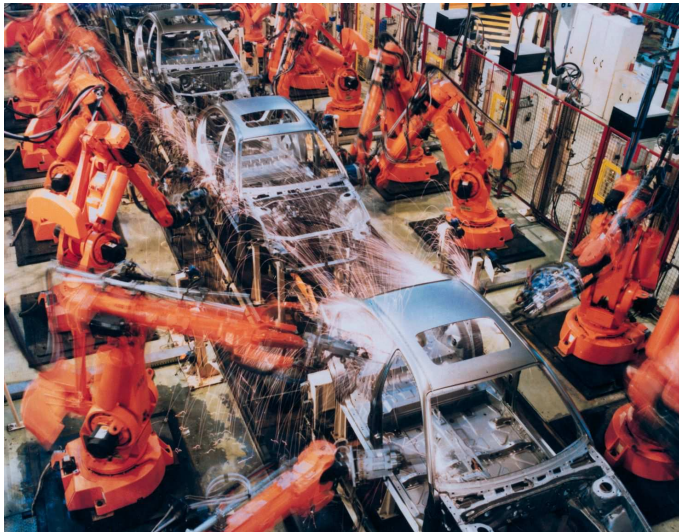
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Robotics: Background



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Definitions

Many definitions have been suggested for what we call a robot.

- 1 The word may conjure up various levels of technological sophistication, ranging from a **simple material handling device** to a **humanoid**.
- 2 The image or the shape of robots varies widely with researchers, engineers, and robot manufacturers.

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Who invented the robot name?

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Devol's industrial robots have their origins in two preceding technologies:

- 1 numerical control for machine tools,

Serial chains of rigid bodies connected by the joints

They are encountered in **industrial robot manipulators**, **robot hands and arms** and **legs of humanoid robots**.

The mathematical description of the serial chain needs the determination of

- **pose (position and orientation)** and **displacement (translation and rotation)** of the robot segments.
- The geometric robot model.

Modern robotics is a branch of science studying the intelligent systems dealing mainly with **movement**, which can be divided into larger groups:

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Movement of various living organisms

Movement invented by humans

There are robotic systems whose movement was invented by humans:

- 1 mobile robots,
- 2 underwater robots,
- 3 flying robots.

mobile robotic systems

We consider autonomous vehicles, predominantly those with wheels.

- vacuum cleaners
- autonomous cars
- industrial mobile robots are used in manufacturing environments for tasks such as **transporting materials**, **handling delicate parts**, bf inspecting products
- Agricultural robots, Construction robots, Disaster response robots
- Service mobile robots include hotel reception robots, household vacuuming robots, restaurant delivery robots

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flying robots robots

autonomous aerial vehicles or autonomous unmanned aerial system (UAS)

Biologically conceived robots

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copy human movements

Most frequent robot systems are robot arms and robot wrists. These are either independent robot manipulators or arms of humanoid robots. Usually they have **six degrees of freedom**.

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copy human movements

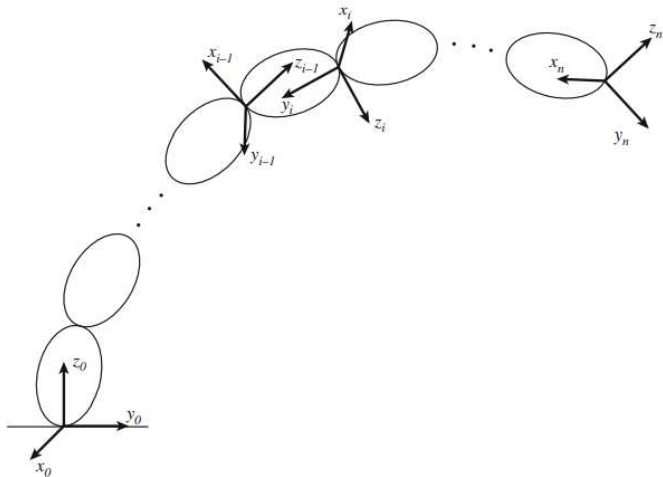
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mechanisms inspired by the world of animals

- robots copying the movements of snakes: Can be employed in inspection and repair in different tubes and funnels
- A lifelike robotic fishs,
- real-life robotic pets
- robots imitating the movements of quadrupeds and other legged insects

serial chain of segments and joints

An open serial chain is shown in Fig. 1. The serial chain is represented by a system of rigid bodies, where each body is connected to two neighboring bodies, except the first and the last body.



Chapter 2: Actuators and Drive Systems

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Actuators are **electro-mechanical** devices that manipulate or move parts of a robot according to controller instructions. A robot has many degrees of freedom (DOF), where each one is a **servoed joint** to generate a desired motion.

To understand the behavior of servoed joints, we consider

- 1 Basic actuator characteristics,
- 2 Drive amplifiers: linear power amplifier and Pulse width modulation (PWM)

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DC motor?

A DC motor is an electrical motor that uses **direct current (DC)** to produce mechanical force. The most common types rely on magnetic forces produced by currents.

DC motors

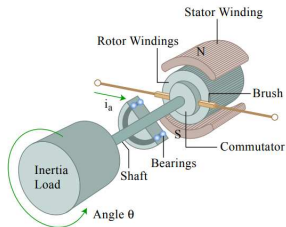


Figure: Components of DC motor

Let us first consider an **idealized energy transducer** having no power loss in converting **electric power** $\Rightarrow$ **mechanical power**. Let E be the voltage applied to the idealized transducer. The electric power $E \cdot i$ must be equivalent to the mechanical power $\tau_m \cdot \omega_m$:

τ_m the torque created at the air gap
 i the current flowing to the rotor windings
 K_t the **torque constant**

The torque is given by

$$\tau_m = K_t \cdot i$$

(1) The voltage E is called the **back emf**

$$P_{in} = E \cdot i = \tau_m \cdot \omega_m \quad (2)$$

$$E = K_t \cdot \omega_m \quad (3)$$

DC Motors

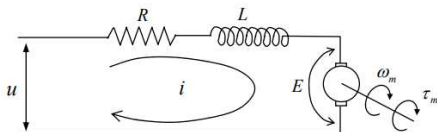


Figure: Electric circuit of armature

But the DC motor **is not a loss-less transducer**, having **resistance** at the the rotor windings and inductance $\Rightarrow$ stores energy.

$$u = R \cdot i + L \frac{di}{dt} + E \quad (4)$$

Combining Eqs. (1), (3) and (4), the mathematical relationship among the applied voltage u , the rotor velocity ω_m , and the motor torque τ_m can be expressed as:

$$\frac{K_t}{R} u = \tau_m + T_e \frac{d\tau_m}{dt} + \frac{K_t^2}{R} \omega_m \quad (5)$$

where the time constant $T_e = \frac{L}{R}$, called the motor reactance, is often negligibly small. Thus, the above equation reduces to an algebraic relationship:

$$\tau_m = \frac{K_t}{R} u - \frac{K_t^2}{R} \omega_m \quad (6)$$

This is called the **torque-speed characteristic**.

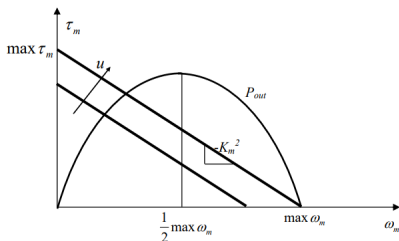


Figure: Torque-speed characteristics and output power

The negative slope of the straight line, $-K_t^2/R$, implies that the voltage-controlled DC motor has an inherent damping in its mechanical behavior.

The power dissipated in the DC motor is given by

$$P_{dis} = R \cdot i^2 = \frac{R}{K_t^2} \tau_m^2 \quad (7)$$

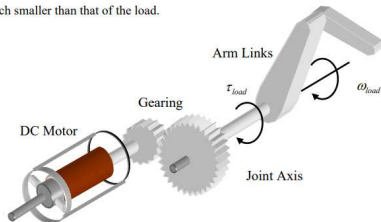
from Eq. (1), we can obtain:

$$\sqrt{P_{dis}} = \frac{\tau_m}{K_m}, \quad K_m = \frac{K_t}{\sqrt{R}} \quad (8)$$

where the parameter K_m is called the motor constant.

Dynamics of single-Axis Drive Systems

is much smaller than that of the load.



Note that the gear reducer of gear ratio r increases the impedance r^2 times larger than that of the motor axis Z_m :

$$Z_{load} = r^2 \cdot Z_m$$

Let I_m be the **inertia** of the motor rotor. From the free body diagram of the motor rotor,

$$I_m \dot{\omega}_m = \tau_m - \frac{1}{r} \tau_{load}$$

$$Z_m = \frac{\text{torque}}{\text{angular velocity}} = \frac{\tau_m}{\omega_m} \quad (9)$$

Let $r > 1$ be a **gear reduction ratio** (If d_1 and d_2 are diameters of two gears, $r = d_2/d_1$. The torque and angular velocity are changed to:

$$\tau_{load} = r \cdot \tau_m, \quad \omega_{load} = \frac{1}{r} \omega_m \quad (10)$$

where $-\frac{1}{r} \tau_{load}$ is the torque acting on the motor shaft from the joint axis through the gears, and $\dot{\omega}_m$ is the time rate of change of angular velocity, i.e. the angular acceleration.

Dynamics of single-Axis Drive Systems

Let I_l be the **inertia of the arm link**, and b the **damping coefficient** of the bearings supporting the joint axis. Considering the free body diagram of the arm link and joint axis yields

$$I_l \dot{\omega}_{load} = \tau_{load} - b \omega_{load}$$

Eliminating τ_{load} from the above two equations and using Eq. (6) and (10) yields

$$I_l \dot{\omega}_{load} + B \omega_{load} = k \cdot u$$

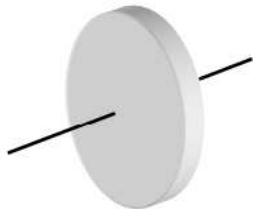
where I , B , k are the **effective inertia, damping, and input gain** reflected to the joint axis:

$$\begin{aligned} I &= I_l + r^2 I_m \\ B &= b + r^2 K_m^2 \\ k &= r \frac{K_t}{R} \end{aligned}$$

Dynamics of single-Axis Drive Systems



(a) Long and slender



(b) Pancake

Figure: Two ways of reducing the motor rotor inertia

Pulse width modulation (PWM)

Actuators are used to control precisely to reach the **desired motions** of arms and legs, which requires **power amplifier** to drive a desired level of voltage (or current indirectly) to the motor armature.

● linear amplifier.

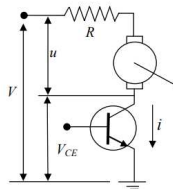


Figure: Analogue power amplifier for driving the armature voltage

Consider the simple circuit of fig. 6 to control the armature voltage with a single transistor.

Let V be the supply voltage connected to one end of the motor armature, and the other end of the armature is connected to the collector of the transistor.

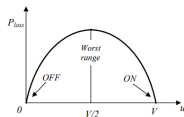


Figure: Power loss at the transistor vs. the armature voltage

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- 2 Pulse Width Modulation or PWM, is the most commonly used method for varying the **average voltage** to the motor.

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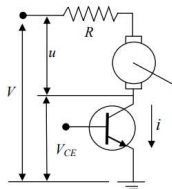


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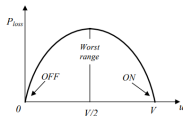


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Pulse width modulation (PWM)

- As shown in Fig. 7, it is clear that the **heat loss** is zero when the armature voltage is either **0** or **V**. This means that the transistor is completely shutting down the current (**OFF**) or completely admitting the current (**ON**). PWM varies the ratio of time length of the complete ON state to the complete OFF state.

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A single cycle of ON and OFF states is called the **PWM period**

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PWM duty cycle

the percentage of the ON state in a single period is called the **duty cycle**.

PWM illustration

Fig. 8 illustrates PWM signals.

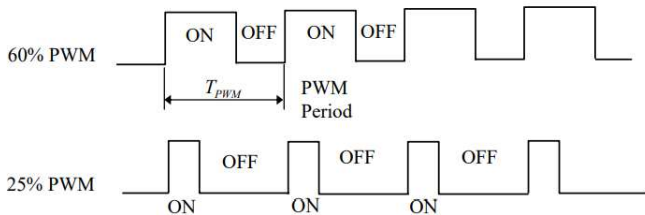


Figure: Pulse width modulation

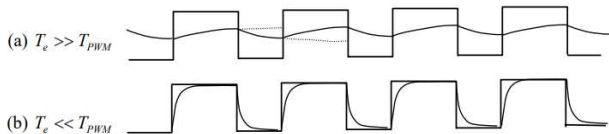
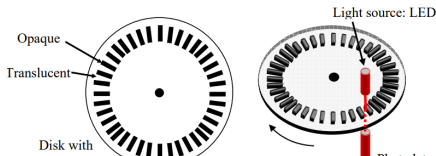


Figure: Current to the motor is smoothed due to inductance

Optical Shaft Encoders

The basic servomechanism based on analogue feedback technology, using potentiometers and tachometer generators are very simple but having limited reliability and performance.

Nowadays, the optical shaft encoders are widely used.



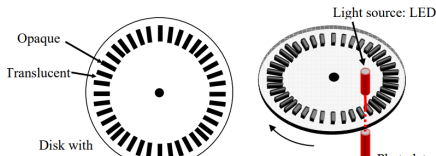
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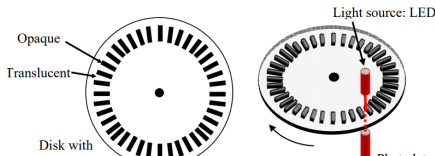
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Position measurement

Fig 11 represents a double track pattern and resultant output signals for clockwise and counter clockwise rotations.



Note that the basic double track encoder requires initialization of the counter before the actual measurement, which is called **incremental encoder**.

An **absolute encoder** gives an n -bit absolute position as well as the direction of rotation without use of an up-down counter and initialization.

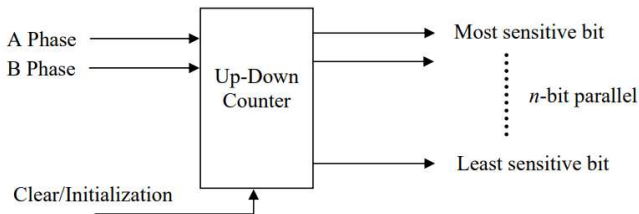


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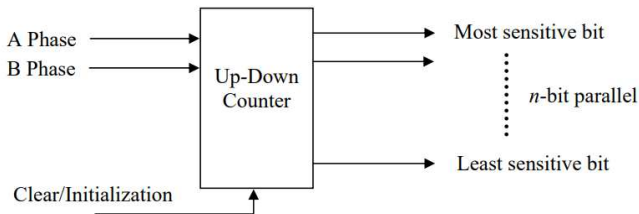


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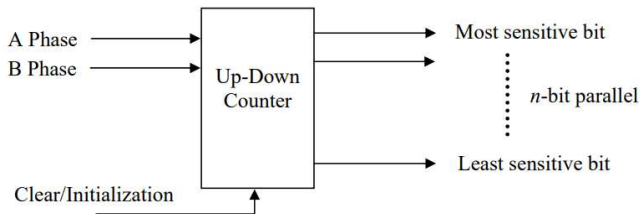
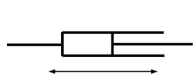


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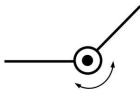
Velocity measurement

Fig 13 represents a pulse sequence coming from an optical encoder. Each pulse indicates a rising edge or a falling edge of phase A and B signals. Thus, the angular velocity can be estimated from the density of the pulse train, i.e. the pulse frequency in every fixed period, for example $T=10$ ms.

Chapter 3: Robot Mechanisms



(a) Prismatic joint



(b) Revolute joint

Figure: Basic joint types.

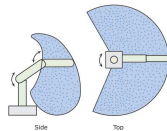
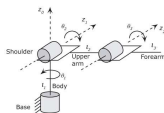


Figure: Articulated robot.

Chap 4: Kinematics

Kinematics?

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Planar Kinematics?

Robot mechanisms that work within a plane, i.e *Planar Kinematics* are simpler than the general **three-dimensional** kinematics. However, most of the robot mechanisms of practical importance can't be treated as planar mechanisms, or reduced to planar problems.

General three-dimensional kinematics needs special mathematical formulation and tools.

Planar Kinematics of Serial Link Mechanisms

Consider the 3 DOF planar arm, Fig. 17. The arm consists of one **fixed link** and 3 **movable links** that moves within the plane.

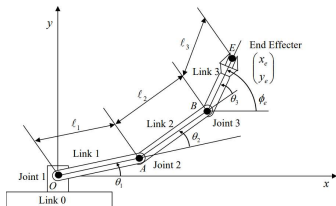


Figure: 3 DOF (degree-of-freedom) robot with 3 revolute joints

Only few geometric parameters are required to describe this robot arm.

Let O , A , and B be the locations of the 3 joint axes, and point E be a point fixed to the **end-effector**. The link lengths are $\ell_1 =$

Actuator 1 driving link 1 is fixed to base link (link 0) with angle θ_1 , while actuator 2 driving link 2 is fixed to the tip of Link 1 with angle θ_2 between the two links, and actuator 3 driving link 3 is fixed to the tip of Link 2 with angle θ_3 . We are concerned with the **location of the end-effector**, which can be described by using a coordinate system $O - xy$, fixed to the base link y_e :

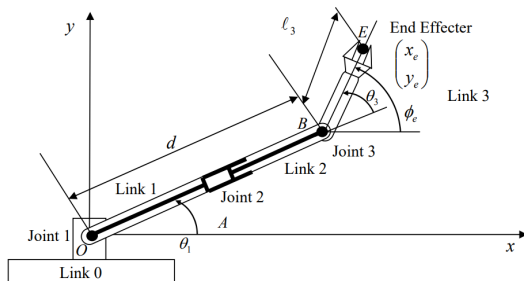
$$\begin{aligned} x_e &= \ell_1 \cos \theta_1 + \ell_2 \cos(\theta_1 + \theta_2) \\ &\quad + \ell_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ y_e &= \ell_1 \sin \theta_1 + \ell_2 \sin(\theta_1 + \theta_2) \\ &\quad + \ell_3 \sin(\theta_1 + \theta_2 + \theta_3) \end{aligned} \quad (11)$$

The **orientation of the end-effector** ϕ_e is related to the **actuator displacements** as:

$$\phi_e = \theta_1 + \theta_2 + \theta_3 \quad (12)$$

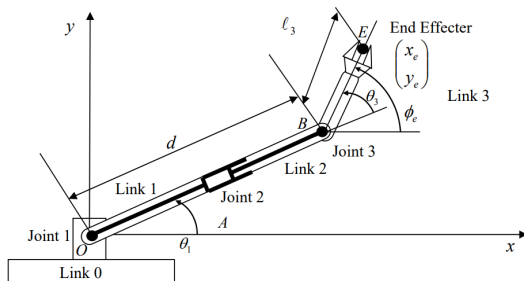
The above equations describe the **position** and **orientation** of the robot end-effector viewed from the **fixed coordinate system in relation to the actuator displacements**.

Exercise: Using the geometric parameters and joint displacement, obtain the kinematic equations relating the end-effector position and the orientation to the joint displacements.



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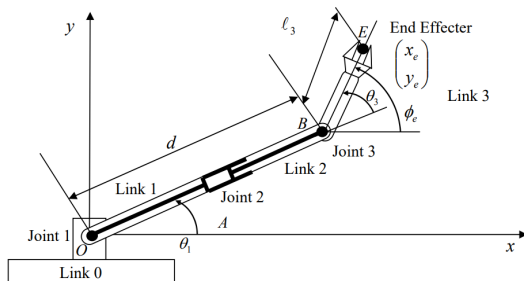
Kinematic model

In general, a set of *algebraic equations* relating the position and orientation of a

- robot end-effector,
- or any significant part of the robot,

to actuator or active joint displacements, is called **Kinematic Equations**, or more specifically, Forward Kinematic Equations.

Exercise: Using the geometric parameters and joint displacement, obtain the kinematic equations relating the end-effector position and the orientation to the joint displacements.



Kinematic equations

There are two types of joints, **prismatic** and **revolute joints**, forming robot mechanisms.. The displacement of the i -th joint is described by

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Let $q = [q_1 \quad q_2 \quad \cdots \quad q_n]^T$ represent all the joint displacements involved in a robot mechanism, where n is the number of joints. **Kinematic equations relate these joint displacements to the position and orientation of the end-effector.** Let us denote the end-effector position and orientation by vector p , which is described by three variables for planar mechanisms.

$$p = \begin{bmatrix} x_e \\ y_e \\ \phi_e \end{bmatrix}$$

Kinematic equations

The kinematic equations are:

$$p = f(q), \quad p \in R^{3 \times 1}, \quad q \in R^{n \times 1} \quad (14)$$

- For a serial a serial link mechanism, all joints are usually **active joints** driven by individual actuators, which uniquely determine the end-effector position en orientation.

The collection of the joint displacements, i.e. the vector **q**, uniquely determine the entire robot configuration.

The joint displacements are **generalized coordinates** that locate the robot mechanism uniquely and completely, and their number is called **degree of freedom**. Vector **q** is called **joint coordinates**, forming a complete and independent set of generalized coordinates.

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- For a serial a serial link mechanism, all joints are usually **active joints** driven by individual actuators, which uniquely determine the end-effector position en orientation.
- If there is a link whose location is not fully determined by the actuator displacements, such a robot is said to be **under-actuated**.

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Inverse Kinematics of Planar Mechanisms

Inverse Kinematics of Planar Mechanisms (1)

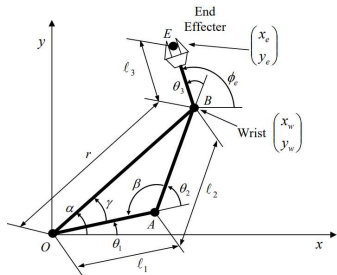
Example 2: Solving the **inverse kinetics** problem is to find the 3 joint angles θ_1 , θ_2 , θ_3 that lead to the **end-effector** to a desired position and orientation, x_e , y_e , ϕ_e . We consider a two-step procedure. First, we find the position of the wrist, point B, from x_e , y_e , ϕ_e . Then, we find θ_1 , θ_2 from the wrist position. Angle θ_3 can be determined directly from the wrist position.

Let x_w and y_w be the coordinates of the wrist. With a motion in the **opposite direction** (to be explained later) to the end-effector orientation ϕ_e , the wrist coordinates are given by

$$\begin{aligned} x_w &= x_e - l_3 \cos(\phi_e) \\ y_w &= y_e - l_3 \sin(\phi_e) \end{aligned} \quad (16)$$

Right hand of this Eq. $\rightarrow$ are function of x_e , y_e , ϕ_e alone. So, we can determine :

$$\alpha = \tan^{-1} (y_w / x_w)$$



Next, let us consider the triangle OAB, which is formed by the wrist B, the elbow A and the shoulder O, and define angles β , γ . Applying the law of cosines to the elbow angle β

$$l_1^2 + l_2^2 - 2l_1 l_2 \cos \beta = r^2$$

where $r^2 = x_w^2 + y_w^2$

$$\theta_2 = \pi - \beta = \pi - \cos^{-1} \left(\frac{l_1^2 + l_2^2 - x_w^2 - y_w^2}{2l_1 l_2} \right)$$

Similarly,

$$r^2 + l_1^2 - 2rl_1 \cos \gamma = l_2^2$$

Solving for γ yields

$$\theta_1 = \alpha - \gamma = \tan^{-1} \left(y_w / x_w \right) - \cos^{-1} \left(\frac{x_w^2 + y_w^2 + l_1^2 - l_2^2}{2l_1 \sqrt{x_w^2 + y_w^2}} \right)$$

From the above θ_1 , θ_2 we can obtain

$$\theta_3 = \phi_e - \theta_1 - \theta_2$$

Kinematics of Parallel Link Mechanisms

Example 3 Consider the 5-bar-kink planar robot arm shown in Figure 20.

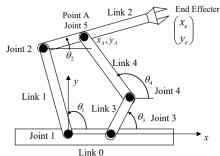


Figure: Multiple solutions to the inverse kinematics of Example 2

$$\begin{aligned}x_e &= l_1 \cos \theta_1 + l_2 \cos \theta_2 \\y_e &= l_1 \sin \theta_1 + l_2 \sin \theta_2\end{aligned}$$

Note that the joint 2 is a **passive joint**. Hence, the angle θ_2 is a dependent variable. Using θ_2 , we can obtain the coordinates of point A:

$$\begin{aligned}x_A &= l_1 \cos \theta_1 + l_5 \cos \theta_2 \\y_A &= l_1 \sin \theta_1 + l_5 \sin \theta_2\end{aligned}$$

Point A must be reached via the branches comprising Links 3 and 4. Therefore,

$$\begin{aligned}x_A &= l_3 \cos \theta_3 + l_4 \cos \theta_4 \\y_A &= l_3 \sin \theta_3 + l_4 \sin \theta_4\end{aligned}$$

Equating these two sets of equations yields two **constraint equations**:

$$\begin{aligned}l_1 \cos \theta_1 + l_5 \cos \theta_2 &= l_3 \cos \theta_3 + l_4 \cos \theta_4 \\l_1 \sin \theta_1 + l_5 \sin \theta_2 &= l_3 \sin \theta_3 + l_4 \sin \theta_4\end{aligned}$$

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Basic 3D geometry

$\varnothing$

Pose (position and orientation)

Frame and coordinate transformns

Quaternion: Homogeneous transformations

Joint types, Kinematic Map, Jacobian

